

AIAC ASSIGNMENT - 16.1

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Task1: Design a database schema for a Student Information System and generate queries using AI.

1 --Task; Design a database schema for a Student Information System and generate queries using AI. 2 -- Or-OP tables already exist (to avoid errors)

3 DROP TABLE IF EXISTS Enrollments;

```
7 -- Students Table
8 CREATE TABLE Students (
9     student_id INTEGER PRIMARY KEY AUTOINCREMENT,
10    first_name TEXT NOT NULL,
11    last_name TEXT NOT NULL,
12    date_of_birth TEXT NOT NULL,
13    email TEXT UNIQUE NOT NULL
14 );
15
16 -- Courses Table
17 CREATE TABLE Courses (
18     course_id INTEGER PRIMARY KEY AUTOINCREMENT,
19     course_name TEXT NOT NULL,
20     course_code TEXT UNIQUE NOT NULL,
21     credits INTEGER NOT NULL
22 );
23
24 -- Enrollments Table
25 CREATE TABLE Enrollments (
26     enrollment_id INTEGER PRIMARY KEY AUTOINCREMENT,
27     student_id INTEGER NOT NULL,
28     course_id INTEGER NOT NULL,
29     enrollment_date TEXT NOT NULL,
30     FOREIGN KEY (student_id) REFERENCES Students(student_id),
31     FOREIGN KEY (course_id) REFERENCES Courses(course_id)
32 );
33
```

4 DROP TABLE IF EXISTS students; 5 DROP TABLE IF EXISTS Courses;



1. Query : Insert a new student record

```
-- Enrollments Table
24 CREATE TABLE Enrollments (
25     enrollment_id INTEGER PRIMARY KEY AUTOINCREMENT,
26     student_id INTEGER NOT NULL,
27     course_id INTEGER NOT NULL,
28     enrollment_date TEXT NOT NULL,
29     FOREIGN KEY (student_id) REFERENCES Students(student_id),
30     FOREIGN KEY (course_id) REFERENCES Courses(course_id)
31 );
32
33 -- Insert Students
34 INSERT INTO Students (first_name, last_name, date_of_birth, email)
35 VALUES
36 ('John', 'Doe', '2000-01-15', 'john@example.com'),
37 ('Jane', 'Smith', '1999-05-20', 'jane@example.com');
38
39 -- Insert Courses
40 INSERT INTO Courses (course_name, credits)
41 VALUES
42 ('Introduction to Computer Science', 3),
43 ('Database Systems', 4);
44
45 -- Enrollments
46 INSERT INTO Enrollments (student_id, course_id, enrollment_date)
47 VALUES
48 (1, 1, '2024-01-10'),
49 (1, 2, '2024-01-12'),
50 (2, 1, '2024-01-11');
```

2. Query : Fetch all courses enrolled by a student.

```
-- Insert Students
34 INSERT INTO Students (first_name, last_name, date_of_birth, email)
35 VALUES
36 ('John', 'Doe', '2000-01-15', 'john@example.com'),
37 ('Jane', 'Smith', '1999-05-20', 'jane@example.com');
38
39 -- Insert Courses
40 INSERT INTO Courses (course_name, course_code, credits)
41 VALUES
42 ('Introduction to Computer Science', 'CS101', 3),
43 ('Database Systems', 'CS102', 4);
44
45 -- Enrollments
46 INSERT INTO Enrollments (student_id, course_id, enrollment_date)
47 VALUES
48 (1, 1, '2024-01-10'),
49 (1, 2, '2024-01-12'),
50 (2, 1, '2024-01-11');
51
52 -- Query
53 SELECT s.first_name, s.last_name, c.course_name
54 FROM Students s
55 JOIN Enrollments e ON s.student_id = e.student_id
56 JOIN Courses c ON e.course_id = c.course_id
57 WHERE c.course_code = 'CS101';
```

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SQL ▾	< 1 / 1 >	1 - 0 of 0	🔍 ↺ ↻ ⌂
SQL ▾	< 1 / 1 >	1 - 0 of 0	🔍 ↺ ↻ ⌂
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first_name	last_name	course_name	
John	Doe	Introduction to Computer Science	
Jane	Smith	Introduction to Computer Science	

3. Query : Count number of students in each course.

```
-- Count number of students In each Course.
SELECT c.Course_name, COUNT(e.student_id) AS student_count
FROM Courses c
LEFT JOIN Enrollments e ON c.course_id = e.course_id
GROUP BY c.course_id;
```

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course_name	student_count		
Introduction to Computer Science	2		
Database Systems	1		

TASK 2 : Create schema and queries for a Hospital Management System.

- Create schema and queries for a Hospital management System

-- Drop tables if already exist (to avoid errors)

DROP TABLE IF EXISTS Appointments;

DROP TABLE IF EXISTS Patients;

DROP TABLE IF EXISTS Doctors; -- Patients Table

```
CREATE TABLE Patients (
    patient_id INTEGER PRIMARY
    KEY AUTOINCREMENT, first_name TEXT NOT NULL, last_name
    TEXT NOT NULL, date_of_birth TEXT NOT NULL, email TEXT
    UNIQUE NOT NULL
)
```

-- Doctors Table

```
CREATE TABLE Doctors (
    doctor_id INTEGER PRIMARY KEY AUTOINCREMENT, first_name TEXT NOT NULL, last_name TEXT NOT
    NULL, specialty TEXT NOT NULL, email TEXT UNIQUE NOT NULL
)
```

-- Appointments Table

```
CREATE TABLE Appointments (
    appointment_id INTEGER
    PRIMARY KEY AUTOINCREMENT, patient_id INTEGER NOT
    NULL, doctor_id INTEGER NOT NULL, appointment_date TEXT NOT
    NULL,
    FOREIGN KEY (patient_id) REFERENCES Patients(patient
    id), FOREIGN KEY (doctor_id) REFERENCES
    Doctors(doctor_id)
)
```

-- Appointments Table

```
CREATE TABLE Appointments (
    appointment_id INTEGER PRIMARY KEY AUTOINCREMENT, patient_id INTEGER NOT NULL,
    doctor_id INTEGER NOT NULL, appointment_date TEXT NOT NULL,
    FOREIGN KEY (patient_id) REFERENCES patients
    (patient_id), FOREIGN KEY (doctor_id) REFERENCES
    Doctors(doctor_id)
)
```

-- Insert Patients

```
INSERT INTO patients (first_name, last_name, date_of_birth, email)
VALUES
```

```
(
    'Michael', 'Clark', '1990-11-01', 'michael@example.com'
),
(
    'Sarah', 'Johnson', '1985-05-15', 'sarah.johnson@example.com'
);
```

-- Insert Doctors

```
INSERT INTO Doctors (first_name, last_name, specialty, email) VALUES
(
    'Dr. Sarah', 'Johnson', 'Cardiology', 'sarah.johnson@example.com'
),
(
    'Dr. David', 'Smith', 'Neurology', 'david.smith@example.com'
);
```

-- Insert Appointments

```
INSERT INTO Appointments (patient_id, doctor_id, appointment_date)
VALUES
(1, 1, '2024-07-01'),
(1, 2, '2024-07-15'),
(2, 1, '2024-07-10');
```

Query 1: List all appointments for a specific doctor.

```
-- Insert Patients
INSERT INTO Patients (first_name, last_name, date_of_birth, email)
VALUES
('Emily', 'Clark', '1985-07-12', 'emily@example.com'),
('Michael', 'Brown', '1990-11-05', 'michael@example.com');
-- Insert Doctors
INSERT INTO Doctors (first_name, last_name, specialty, email)
VALUES
('Dr. Sarah', 'Johnson', 'Cardiology', 'sarah.johnson@example.com'),
('Dr. David', 'Smith', 'Neurology', 'david.smith@example.com');
-- Insert Appointments
INSERT INTO Appointments (patient_id, doctor_id, appointment_date)
VALUES
(1, 1, '2024-07-01'),
(1, 2, '2024-07-15'),
(2, 1, '2024-07-10');
```

```
-- List all appointments for a specific doctor.
SELECT p.first_name, p.last_name, a.appointment_date
FROM Patients p
JOIN Appointments a ON p.patient_id = a.patient_id
JOIN Doctors d ON a.doctor_id = d.doctor_id
WHERE d.email = 'sarah.johnson@example.com';
```

first_name	last_name	appointment_date
Emily	Clark	2024-07-01
Michael	Brown	2024-07-10

Query 2 : Retrieve patient history by patient ID.

```
-- List all appointments for a specific doctor.
SELECT p.first_name, p.last_name, a.appointment_date
FROM Patients p
JOIN Appointments a ON p.patient_id = a.patient_id
JOIN Doctors d ON a.doctor_id = d.doctor_id
WHERE d.email = 'sarah.johnson@example.com';

-- Retrieve patient history by patient ID.
SELECT d.first_name, d.last_name, a.appointment_date
FROM Doctors d
JOIN Appointments a ON d.doctor_id = a.doctor_id
JOIN Patients p ON a.patient_id = p.patient_id
WHERE p.patient_id = 1;
```

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first_name	last_name	appointment_date	
Emily	Clark	2024-07-01	
Michael	Brown	2024-07-10	

SQL ▾	< 1 / 1 >	1 - 2 of 2	🔍 ↺ ⌂
first_name	last_name	appointment_date	
Dr. Sarah	Johnson	2024-07-01	
Dr. David	Smith	2024-07-15	

Query 3 : Count total patients treated by each doctor.

```
-- Count total patients treated by each doctor.
SELECT d.first_name, d.last_name, COUNT(a.patient_id) AS patient_count
FROM Doctors d
LEFT JOIN Appointments a ON d.doctor_id = a.doctor_id
GROUP BY d.doctor_id;
```

SQL ▾	< 1 / 1 >	1 - 2 of 2	🔍 ↺ ⌂
first_name	last_name	patient_count	
Dr. Sarah	Johnson	2	
Dr. David	Smith	1	

TASK 3 : Design schema for a Library Management System

→ Design schema for a Library Management System

→ Drop tables if already exist (to avoid errors)

DROP TABLE IF EXISTS Borrowings;

OR DROP TABLE IF EXISTS Books;

OR DROP TABLE IF EXISTS Members;

→ Members Table

```
CREATE TABLE Members (
    member_id INTEGER PRIMARY KEY
    AUTOINCREMENT, first_name TEXT NOT NULL,
    last_name TEXT NOT NULL, email TEXT
);
```

→ Books Table

```
CREATE TABLE Books (
    book_id INTEGER PRIMARY KEY
    AUTOINCREMENT, title TEXT NOT NULL,
    author TEXT NOT NULL,
    isbn TEXT UNIQUE
);
```

→ Borrowings Table

```
CREATE TABLE Borrowings (
    borrowing_id INTEGER PRIMARY KEY
    AUTOINCREMENT, member_id INTEGER NOT NULL,
```

```

book_id INTEGER NOT NULL, borrow_date TEXT
NULL, return_date TEXT,
FOREIGN KEY REFERENCES Members(member_id),
FOREIGN KEY (book_id) REFERENCES Books(book_id)
REFERENCES

```

```

-- Insert members
INSERT INTO Members (first_name, last_name, email)
VALUES ('Alice', 'Johnson', 'alice@example.com'),
       ('Bob', 'Smith', 'bob@example.com');

-- Insert Books
INSERT INTO Books (title, author, isbn)
VALUES ('The Great Gatsby', 'F. Scott Fitzgerald', '978-0-7432-7356-5'),
       ('To Kill a Mockingbird', 'Harper Lee', '978-0-06-114003-6');

-- Insert Borrowings
INSERT INTO Borrowings (member_id, book_id, borrow_date, return_date)
VALUES (1, 1, '2024-08-01', '2024-08-15'),
       (1, 2, '2024-08-16', NULL),
       (2, 1, '2024-08-05', '2024-08-20');

INSERT INTO Books (title, author, isbn)
VALUES

```

Query 1 : Retrieve all books currently issued.

```

-- Retrieve all books currently issued.
SELECT b.title, m.first_name, m.last_name, br.borrow_date
FROM Books b
JOIN Borrowings br ON b.book_id = br.book_id
JOIN Members m ON br.member_id = m.member_id
WHERE br.return_date IS NULL;

```

SQL

title	first name	last name	date
To Kill a Mockingbird	Alice	Johnson	2024-08-16

Query 2 : Find the most purchased product.

```

-- Retrieve all books currently issued.
SELECT b.title, m.first_name, m.last_name, br.borrow_date
FROM Books b
JOIN Borrowings br ON b.book_id = br.book_id
JOIN Members m ON br.member_id = m.member_id
WHERE br.return_date IS NULL;

-- Find the most purchased product.
SELECT b.title, COUNT(br.borrowing_id) AS borrow_count
FROM Books b
JOIN Borrowings br ON b.book_id = br.book_id
GROUP BY b.book_id
ORDER BY borrow_count DESC
LIMIT 1;

```

title	first_name	last_name	borrow_date
To Kill a Mockingbird	Alice	Johnson	2024-08-16

title	borrow_count
The Great Gatsby	2

Query 3 : Count number of books loaned by each member.

```

-- Find the most purchased product.
SELECT b.title, COUNT(br.borrowing_id) AS borrow_count
FROM Books b
JOIN Borrowings br ON b.book_id = br.book_id
GROUP BY b.book_id
ORDER BY borrow_count DESC
LIMIT 1;

-- Count number of books loaned by each member.
SELECT m.first_name, m.last_name, COUNT(br.borrowing_id) AS borrow_count
FROM Members m
LEFT JOIN Borrowings br ON m.member_id = br.member_id
GROUP BY m.member_id;

```

To Kill a Mockingbird	Alice	Johnson	2024-08-16
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title	borrow_count
The Great Gatsby	2

first_name	last_name	borrow_count
Alice	Johnson	2
Bob	Smith	1

TASK 4 : Design a database for an E-commerce platform.

- Design a database for an E-commerce platform,
- Drop tables if already exist (to avoid errors)


```
DROP IF EXISTS Orders;
DROP TABLE IF EXISTS Products;
DROP TABLE IF EXISTS customers;
```

```
-- Customers Table
CREATE TABLE customers ( customer_id INTEGER
    PRIMARY KEY AUTOINCREMENT, first_name
    TEXT NOT NULL,
    last_name TEXT NOT NULL, email
    TEXT NOT NULL);
```

```
Products Table
CREATE TABLE products ( product_id PRIMARY KEY
    AUTOINCREMENT, product_name TEXT
    NOT NULL, price REAL NOT NULL, stock
    INTEGER NOT NULL);
```

```
-- Orders Table
CREATE TABLE orders (order_id INTEGER PRIMARY
    KEY AUTOINCREMENT, customer_id INTEGER NOT
    NULL, product_id INTEGER NOT NULL, order_date
    TEXT NOT NULL, quantity INTEGER
    NOT NULL,
    FOREIGN KEY (customer_id) REFERENCES customers(customer_id),
    FOREIGN KEY (product_id) REFERENCES
```

```
Products(product_id))
```

```
-- Insert Customers
```

```
INSERT INTO Customers (first_name, last_name, email)
VALUES
    ('John', 'Doe', 'john@example.com'),
    ('Jane', 'Smith', 'jane@example.com');
```

```
-- Insert Products
```

```
INSERT INTO Products (product_name, price, stock)
VALUES ('Laptop',
    999.99, 10),
    ('Mouse', 29.99, 50);
-- Insert Orders
INSERT INTO Orders (customer_id, product_id, order_date, quantity)
VALUES
    (1, 1, '2024-08-01', 1),
    (1, 2, '2024-08-01', 2),
    (2, 1, '2024-08-02', 1);
```

Query 1 : Retrieve all orders by a user.

```
-- Retrieve all orders by a user.
SELECT c.first_name, c.last_name, p.product_name, o.order_date, o
    quantity
FROM Customers c
JOIN Orders o ON c.customer_id = o.customer_id
JOIN Products p ON o.product_id = p.product_id
WHERE c.email = 'john@example.com';
```

first_name	last_name	product_name	order_date	quantity
John	Doe	Laptop	2024-08-01	1
John	Doe	Mouse	2024-08-01	2

2 : Find the most purchased product. Query

```
-- Retrieve all orders by a user.
SELECT c.first_name, c.last_name, p.product_name, o.order_date, o
    quantity
FROM Customers c
JOIN Orders o ON c.customer_id = o.customer_id
JOIN Products p ON o.product_id = p.product_id
WHERE c.email = 'john@example.com';

-- Find the most purchased product.
SELECT p.product_name, SUM(o.quantity) AS total_quantity
FROM Products p
JOIN Orders o ON p.product_id = o.product_id
GROUP BY p.product_id
ORDER BY total_quantity DESC
LIMIT 1;
```

first_name	last_name	product_name	order_date	quantity
John	Doe	Laptop	2024-08-01	1
John	Doe	Mouse	2024-08-01	2

product_name	total_quantity
Laptop	2

Query 3 : Calculate total revenue in a given month

```
-- Find the most purchased product.
SELECT p.product_name, SUM(o.quantity) AS total_quantity
FROM Products p
JOIN Orders o ON p.product_id = o.product_id
GROUP BY p.product_id
ORDER BY total_quantity DESC
```

product_name	total_quantity
Laptop	2

```
-- Calculate total revenue in a given month
SELECT strftime('%Y-%m', o.order_date) AS month, SUM(p.price * o.quantity) AS total_revenue
FROM Orders o
JOIN Products p ON o.product_id = p.product_id
GROUP BY month
ORDER BY month;
```

month	total_revenue
2024-08	2059.96

1 - 1 of 1

TASK 5 : Design a database schema for a University Examination System and generate SQL queries using AI.

- Design a database schema for university Examination System and generate SQL queries using AI. - Orlop tables if already exist (to avoid

OROP

DROP TABLE IF EXISTS students;

DROP TABLE IF EXISTS Exams; --

Students Table

```
CREATE TABLE students ( student_id INTEGER PRIMARY KEY
AUTOINCREMENT, first_name TEXT NULL, last name
```

```
last_name TEXT NOT NULL,
date_of_birth TEXT NOT NULL,
email TEXT UNIQUE NOT NULL
);
```

-- Exams Table

```
CREATE TABLE Exams (
exam_id INTEGER PRIMARY KEY AUTOINCREMENT,
course_name TEXT NOT NULL,
exam_date TEXT NOT NULL
);
```

-- ExamResults Table

```
CREATE TABLE ExamResults (
result_id INTEGER PRIMARY KEY
student_id INTEGER NOT NULL,
exam_id INTEGER NOT NULL,
score REAL NOT NULL,
FOREIGN KEY (student_id) REFERENCES students(student_id),
FOREIGN KEY (exam_id) REFERENCES Exams(exam_id)
);
```

```
CREATE TABLE Exams ( exam_id INTEGER PRIMARY KEY
AUTOINCREMENT, course_name TEXT NULL,
exam_date TEXT NULL
);
```

-- ExamResults Table

```
CREATE TABLE ExamResults ( result_id INTEGER PRIMARY KEY
AUTOINCREMENT,
```

```
student_id INTEGER NOT NULL,
exam_id INTEGER NOT NULL,
score REAL NOT NULL,
FOREIGN KEY (student_id) REFERENCES students(student_id),
FOREIGN KEY (exam_id) REFERENCES Exams(exam_id)
);
```

-- Insert Students

```
INSERT INTO students (first_name, last_name, date_of_birth, email)
VALUES
```

```
( 'Alice', 'Johnson', '2003-03-22', 'alice@example.com' ),
```

```
( 'Bob', 'Smith', '2000-07-15', 'bob@example.com');
-- Insert
Exams
INSERT INTO Exams (course_name, exam_date)
VALUES
    ('2024-08-01', (
'Physics'.
    Insert Exam Result5
        (student_id, exam_id score)
VALUES
(1, 1, 85.5).
(1.
    g2.
ey.
(2, 1, 78.5),
```

Query 1 : Insert examination results for a student.

```
-- Query 1 : Insert examination results for a student.
INSERT INTO ExamResults (student_id, exam_id, score)
VALUES (1, 1, 85.5);
```

Query 2: Retrieve subject-wise marks for a student.

```
-- Query 2: Retrieve subject-wise marks for a student.
SELECT e.course_name, er.score
FROM ExamResults er
JOIN Exams e ON er.exam_id = e.exam_id
WHERE er.student_id = 1;
```

course_name	Score
"athenatics5	85.5
"athematics	85.5

Query 3 : Calculate average marks for each subject.

```
-- Query 3: Calculate the average score for a specific exam.
SELECT e.course_name, AVG(er.score) AS average_score
FROM ExamResults er
JOIN Exams e ON er.exam_id = e.exam_id
WHERE e.course_name = 'Mathematics'
GROUP BY e.exam_id;
```

Mathematics	85.5
-------------	------

course_name	average_score
Mathematics	83.0

1 - 1 0+1

LAB OBSERVATIONS

◆ Objective

To design and implement database schemas for various real-world systems and perform SQL operations such as table creation, data insertion, and query execution using SQL.

Observation 1: Student Information System

- ◆ Created three tables: Students, Courses, Enrollments.
- ◆ Established many-to-many relationship using a bridge table (Enrollments).

- ◆ Used primary keys and foreign keys to maintain data integrity.
- Inserted sample student and course data successfully.
- Performed queries to:
 - ◆ Retrieve students enrolled in a specific course.
 - ◆ Count number of students in each course.
 - ◆ Observed that JOIN operations help combine data from multiple tables.

Observation 2: Hospital Management System

- Designed tables: Patients, Doctors, Appointments. •
- Implemented relationships using foreign keys. •
- Inserted patient, doctor, and appointment data.

- ◆ Executed queries to:
 - ◆ List appointments for a specific doctor.
 - ◆ Retrieve patient history.
 - ⁰ Count total patients handled by each doctor.

- Observed that LEFT JOIN helps include doctors with no appointments.

Observation 3: Library Management System

- ◆ Created tables: Members, Books, Borrowings. • Maintained relationships using foreign keys.
- ◆ Inserted borrowing records with return status.
- Executed queries to:
 - ◆ Find currently issued books (using NULL condition).
 - ◆ Identify most borrowed book.
 - ◆ Count books borrowed by each member.
 - ◆ Observed that NULL values are useful for tracking incomplete actions (e.g., not returned books).

◆ Observation 4: E-Commerce System

- ◆ Designed tables: Customers, Products, Orders.
- Established relationships using foreign keys.

- ◆ Inserted sample order and product data.
- ◆ Executed queries to:
- ◆ Retrieve all orders by a specific customer.
- ◆ Find most purchased product.
- ◆ Calculate total revenue per month.
- ◆ Observed that aggregate functions (SUM, COUNT) are useful for business analysis.

◆ Observation 5: University Examination System

•Created tables: Students, Exams,ExamResults .

- ◆ Linked students and exams using a result table.
- ◆ Inserted exam and result data.
- ◆ Executed queries to:
- ◆ Retrieve subject-wise marks.
- ◆ Calculate average scores for exams.
- Observed that GROUP BY and AVG functions help in performance evaluation.